

Basics of Computing -- Guided Notes ANSWER KEY

Unit tp-1.1 | CompTIA Network+ N10-009 Obj. FC0-U71 1.1 | N10-008 FC0-U71 1.1

For instructor use / distribute after students complete the notes.

Question 1

A _____ is any physical component of a computer that you can touch, such as a keyboard, CPU, or hard drive. A _____ is a set of instructions that tells hardware what to do, such as an operating system or application.

Answer: Hardware; software.

Question 2

Every computer performs four fundamental functions: _____ (keyboard, mouse), _____ (CPU), _____ (RAM, hard drive), and _____ (monitor, speaker). The CPU coordinates all four of these functions.

Answer: Input; processing; storage; output.

Question 3

Analog signals are _____ (continuous/discrete) -- they can take any value within a range. Digital signals are _____ (continuous/discrete) -- they use only defined states such as 0 and 1. Computers process _____ data because transistors have two stable states: on and off.

Answer: Continuous; discrete; digital.

Question 4

Binary is the language computers use internally because transistors have only two states: _____ (represented as 1) and _____ (represented as 0). Each binary digit is called a _____, meaning Binary digIT.

Answer: On (conducting); off (not conducting); bit.

Question 5

The CPU (Central Processing Unit) is responsible for _____ instructions. It fetches an instruction from memory, _____ it to determine what to do, and then _____ the operation. Modern CPUs perform billions of these cycles per second.

Answer: Executing; decodes; executes.

Question 6

The operating system (OS) is the software that manages _____ resources (CPU, RAM, storage) and provides a platform for _____ to run. Without an OS, application software cannot communicate with the hardware.

Answer: Hardware; application software (apps).

Question 7

In a client-server model, the _____ device makes requests and the _____ device responds with data or services. A web browser on your laptop is an example of a _____, while the machine hosting the website is the _____.

Answer: Client; server; client; server.

Question 8 -- Real World Application

REAL WORLD: A student asks why their smart thermostat is considered a computer even though it does not have a screen or keyboard. Using the four functions of computing, explain how the thermostat qualifies as a computer and give one example of each function in that device.

Answer: The thermostat qualifies as a computer because it performs all four functions. Input: temperature sensors read the current room temperature and the user sets a target temperature via buttons or an app. Processing: the CPU compares the current temperature to the target and decides whether to turn the heating or cooling on. Storage: the thermostat stores the schedule, settings, and historical temperature data. Output: it activates the HVAC system and displays status on a screen or app. Any device that performs all four functions -- regardless of form factor -- is a computer.